Parametric & Sequences Review

worked

Same numbering as the review. Work in blue, the reason in red. Graph choices name the letter on the review.

1. $x = t+1,\ y = 2t^2,\ t$ in $[-1,1]$

$t = x-1,\ \ y = 2(x-1)^2$ — *isolate t in x (one answer), substitute*

t	(x,y)
-1	$(0,2)$
0	$(1,0)$
1	$(2,2)$

$\boxed{y = 2(x-1)^2 \text{ for } x \text{ in } [0,2]}$ — *(a) **B**: opens up, vertex $(1,0)$*

2. $x = \sqrt{t},\ y = 4t-9,\ t$ in $[0,4]$

$x^2 = t,\ \ y = 4x^2-9$ — *square: one answer*

t	(x,y)
0	$(0,-9)$
1	$(1,-5)$
4	$(2,7)$

$\boxed{y = 4x^2-9 \text{ for } x \text{ in } [0,2]}$ — *(a) **A**: right half only, $x \geq 0$*

3. $x = t^3+5,\ y = t^3-2,\ t$ in $(-\infty,\infty)$

$t^3 = x-5,\ \ y = (x-5)-2$ — *solve for t^3, no cube root needed*

$\boxed{y = x-7 \text{ for } x \text{ in } (-\infty,\infty)}$ — *(a) **D**: slope 1, y-intercept -7*

4. $x = 4\sin t,\ y = 4\cos t,\ t$ in $[0,2\pi]$

$\sin t = \frac{x}{4},\ \cos t = \frac{y}{4} \Rightarrow \frac{x^2}{16} + \frac{y^2}{16} = 1$ — $\sin^2 t + \cos^2 t = 1$

$\boxed{x^2+y^2 = 16 \text{ for } x \text{ in } [-4,4]}$ — *(a) **C**: circle, radius 4 (starts at $(0,4)$, clockwise)*

5. $x = 2\tan t,\ y = 3\sec t,\ t$ in $(-\frac{\pi}{2},\frac{\pi}{2})$

$\frac{y^2}{9} - \frac{x^2}{4} = 1$ — $\sec^2 t - \tan^2 t = 1$

$y^2 = 9\left(\frac{x^2}{4}+1\right)$ — *$\sec t > 0$ on this interval: top branch only*

t	(x,y)
$-\frac{\pi}{4}$	$(-2,3\sqrt{2})$
0	$(0,3)$
$\frac{\pi}{4}$	$(2,3\sqrt{2})$

$\boxed{y = 3\sqrt{\frac{x^2}{4}+1} \text{ for } x \text{ in } (-\infty,\infty)}$ — *(a) **C**: top branch with asymptotes*

6. $x = 4\sin t,\ y = 2\csc t,\ t$ in $(0,\pi)$

$\sin t = \frac{x}{4} \Rightarrow \csc t = \frac{4}{x}$ — *reciprocal*

$y = 2 \cdot \frac{4}{x}$ — *substitute*

t	(x,y)
$\frac{\pi}{6}$	$(2,4)$
$\frac{\pi}{2}$	$(4,2)$
$\frac{5\pi}{6}$	$(2,4)$

$\boxed{y = \frac{8}{x} \text{ for } x \text{ in } (0,4]}$ — *$\sin t$ on $(0,\pi)$ is in $(0,1]$: x in $(0,4]$. (a) **D**; it goes down to $(4,2)$ and back (retraced)*

7. $x = t,\ y = \sqrt{t^2+19}$

$\boxed{y = \sqrt{x^2+19} \text{ for } x \text{ in } (-\infty,\infty)}$ — *$t = x$; (a) **D**: one curve above the axis, y-int $\sqrt{19}$*

8. $x = t+2,\ y = \frac{5}{t+2},\ t \neq -2$

$\boxed{y = \frac{5}{x} \text{ for } x \text{ in } (-\infty, 0) \cup (0, \infty)}$ *denominator is x; (a) **D**: quadrants I and III*

9. $x = 17\cos t,\ y = 17\sin t,\ t$ in $[0, 2\pi]$

$\boxed{x^2 + y^2 = 289}$ $\left(\frac{x}{17}\right)^2 + \left(\frac{y}{17}\right)^2 = 1$; *graph **B***

10. $y = (x+4)^2 - 2$

$x = t$: $\boxed{y = (t+4)^2 - 2}$ *do not factor or expand*

$x = t - 4$: $y = (t - 4 + 4)^2 - 2$

$\boxed{y = t^2 - 2}$ *both for t in* $(-\infty, \infty)$

11. $y = x^2 - 4x + 10$

$x = t$: $\boxed{y = t^2 - 4t + 10}$

$x = t + 2$: $(t+2)^2 - 4(t+2) + 10$ *substitute, simplify*

$t^2 + 4t + 4 - 4t - 8 + 10 = \boxed{t^2 + 6}$

12. Baseball: $h = 5$ ft, 22°, 160 ft/s.

(a) **C**: $x = 160\cos(22^\circ)t,\ y = -16t^2 + 160\sin(22^\circ)t + 5$ $v = 160,\ \theta = 22^\circ,\ h = 5$

(b) $t = \frac{x}{160\cos 22^\circ}$ into y ***B***: $y = 5 - \frac{x^2}{1600\cos^2 22^\circ} + (\tan 22^\circ)x$

(c) graph, trace to $y \approx 0$: $\boxed{t \approx 3.83 \text{ s}}$ *degree mode; tmax past 4*

$x \approx 160\cos(22^\circ)(3.83) \approx \boxed{568.18 \text{ ft}}$ *the review rounds t first; unrounded t gives 567.84*

13. Baseball: $h = 3$ ft, 60°, 76 ft/s.

(a) $x = 38t,\ \boxed{y = -16t^2 + 38\sqrt{3}t + 3}$ $76\cos 60^\circ = 38,\ 76\sin 60^\circ = 38\sqrt{3}$

(b) $y \approx 0$ at $t \approx 4.16$: $x \approx 38(4.16) = \boxed{158.1 \text{ ft}}$ *trace to $y \approx 0$*

(c) top at $t \approx 2.06$: $\boxed{70.7 \text{ ft}}$ high, $x \approx \boxed{78.3 \text{ ft}}$ *intermediates to hundredths, as the directions say*

(d) $x = 100 \Rightarrow t = \frac{100}{38} \approx 2.63,\ y \approx 65.4$ ***A**: yes, far above 6 ft*

14. $\left\{\frac{n}{n+6}\right\}$

$\frac{1}{7}, \frac{2}{8}, \frac{3}{9}, \frac{4}{10}, \frac{5}{11} \rightarrow \boxed{\frac{1}{7}, \frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{5}{11}}$

15. $\{(-1)^n n^2\}$

$\boxed{-1,\ 4,\ -9,\ 16,\ -25}$ *odd n negative*

16. $\left\{\frac{2^n}{2^n+9}\right\}$

$\frac{2}{11}, \frac{4}{13}, \frac{8}{17}, \frac{16}{25}, \frac{32}{41}$ *already reduced*

17. $\frac{1}{8}, \frac{2}{9}, \frac{3}{10}, \frac{4}{11}, \ldots$

$\boxed{a_n = \frac{n}{n+7}}$ *top n, bottom 7 more*

18. $1, \frac{1}{9}, \frac{1}{81}, \frac{1}{729}, \ldots$

$\boxed{a_n = \left(\frac{1}{9}\right)^{n-1}}$ *first term 1, so power $n-1$*

19. $-2, 2, -2, 2, \ldots$

$\boxed{a_n = (-1)^n \cdot 2}$ *first term negative: $(-1)^n$*

20. $4, -8, 12, -16, \ldots$

$\boxed{a_n = 4n(-1)^{n+1}}$ *first positive: $n+1$*

21. $a_1 = 7;\ a_n = \frac{a_{n-1}}{n}$

$a_2 = \frac{7}{2},\ a_3 = \frac{7/2}{3} = \frac{7}{6}$ *divide by the term number*

$a_4 = \frac{7}{24},\ a_5 = \frac{7}{120}$

$\boxed{7, \frac{7}{2}, \frac{7}{6}, \frac{7}{24}, \frac{7}{120}}$

22. $a_1 = -2;\ a_2 = 2;\ a_n = a_{n-1} \cdot a_{n-2}$

$a_3 = a_2 a_1 = 2(-2) = -4$ *previous two multiplied*

$a_4 = (-4)(2) = -8,\ \ a_5 = (-8)(-4) = 32$

$\boxed{-2,\ 2,\ -4,\ -8,\ 32}$

23. $\sum_{k=1}^{n}(k+7)$

$\boxed{8+9+\cdots+(n+7)}$ *first $k=1$, second $k=2$, last $k=n$*

24. $\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\cdots+\frac{20}{20+1}$

$\boxed{\sum_{k=1}^{20}\frac{k}{k+1}}$

25. $1-\frac{1}{8}+\frac{1}{64}-\frac{1}{512}+\cdots+(-1)^8\left(\frac{1}{8^8}\right)$

$\boxed{\sum_{k=0}^{8}(-1)^k\left(\frac{1}{8^k}\right)}$ *$k=0$ gives the 1*

26. $\sum_{k=1}^{13} 5$

$5\sum_{k=1}^{13} 1 = 5(13) = \boxed{65}$ *(1), (5)*

27. $\sum_{k=1}^{47} k$

$\frac{47(48)}{2} = \boxed{1128}$ *(6)*

28. $\sum_{k=1}^{5}(3k+9)$

$3\sum_{k=1}^{5} k + 9\sum_{k=1}^{5} 1$ *(2), (1)*

$3\left[\frac{5(6)}{2}\right] + 9(5) = 45+45 =$ *(6), (5)*

$\boxed{90}$

29. $\sum_{k=1}^{3}(k^2-6)$

$\sum_{k=1}^{3} k^2 - 6\sum_{k=1}^{3} 1$ *(3), (1)*

$\frac{3(4)(7)}{6} - 6(3) = 14-18 =$ *(7), (5)*

$\boxed{-4}$

30. $\sum_{k=10}^{30} 8k$

$8\left[\sum_{k=1}^{30} k - \sum_{k=1}^{9} k\right]$ *(1), (4): $j=9$*

$8\left[\frac{30(31)}{2} - \frac{9(10)}{2}\right]$ *(6)*

$8[465-45] = \boxed{3360}$

31. $(x+3)^4$, row 1 4 6 4 1

$1x^4(3)^0 + 4x^3(3)^1 + 6x^2(3)^2 + 4x(3)^3 + 1(3)^4$ *blanks: x down, 3 up*

$\boxed{x^4+12x^3+54x^2+108x+81}$

32. $(x-2)^5$, row 1 5 10 10 5 1, $a=-2$

$x^5+5x^4(-2)+10x^3(-2)^2+10x^2(-2)^3+5x(-2)^4+(-2)^5$ *odd powers of -2 negative*

$\boxed{x^5-10x^4+40x^3-80x^2+80x-32}$

33. $(3p+3)^4$, row 1 4 6 4 1

$(3p)^4+4(3p)^3(3)+6(3p)^2(3)^2+4(3p)(3)^3+3^4$ $(3p)^4=81p^4$*; calculator OK here*

$\boxed{81p^4+324p^3+486p^2+324p+81}$

OPTIONAL Extra Practice: Parametric Equations

worked

1. $x = 6\sin t,\ y = 6\cos t,\ t$ in $[0, 2\pi]$

$\frac{x^2}{36} + \frac{y^2}{36} = 1$ — $\sin^2 t + \cos^2 t = 1$

$\boxed{x^2 + y^2 = 36 \text{ for } x \text{ in } [-6, 6]}$ — *(a)* ***C****: circle, radius 6*

2. $x = 3\tan t,\ y = 4\sec t,\ t$ in $(-\frac{\pi}{2}, \frac{\pi}{2})$

$\frac{y^2}{16} - \frac{x^2}{9} = 1$ — $\sec^2 t - \tan^2 t = 1$

$y^2 = 16\left(\frac{x^2}{9} + 1\right)$ — $\sec t > 0$ *here: top branch only*

$\boxed{y = 4\sqrt{\frac{x^2}{9} + 1} \text{ for } x \text{ in } (-\infty, \infty)}$ — *(a)* ***C****: top branch only*

3. $x = 2\sin t,\ y = 2\csc t,\ t$ in $(0, \pi)$

$\csc t = \frac{2}{x},\ y = 2 \cdot \frac{2}{x}$ — *reciprocal*

$\boxed{y = \frac{4}{x} \text{ for } x \text{ in } (0, 2]}$ — $\sin t$ *on* $(0, \pi)$ *is in* $(0, 1]$*; (a)* ***C****: one piece ending at* $(2, 2)$

4. $x = t,\ y = \sqrt{t^2 + 14}$

$\boxed{y = \sqrt{x^2 + 14} \text{ for } x \text{ in } (-\infty, \infty)}$ — *(a)* ***B****: one curve above the axis*

5. $y = (x+5)^2 - 9$

$x = t$: $\boxed{y = (t+5)^2 - 9}$ — *do not factor*

$x = t - 5$: $\boxed{y = t^2 - 9}$ — *both for* t *in* $(-\infty, \infty)$

6. $y = x^2 - 8x + 19$

$x = t$: $\boxed{y = t^2 - 8t + 19}$

$x = t + 4$: $(t+4)^2 - 8(t+4) + 19$

$= \boxed{t^2 + 3}$

7. Baseball: $h = 3$ ft, 60°, 68 ft/s; 7-ft fence at 100 ft.

(a) $x = 34t$, $\boxed{y = -16t^2 + 34\sqrt{3}t + 3}$ — $68\cos 60° = 34,\ 68\sin 60° = 34\sqrt{3}$

(b) $y \approx 0$ at $t \approx 3.73$: $x \approx 34(3.73) = \boxed{126.8 \text{ ft}}$ — *intermediates to hundredths*

(c) top at $t \approx 1.84$: $\boxed{57.2 \text{ ft}}$ high, $x \approx 34(1.84) = \boxed{62.6 \text{ ft}}$

(d) $t = \frac{100}{34} \approx 2.94,\ y \approx 37.8$ ft — ***B****: yes, higher than 7 ft at 100 ft*